

HW Class 06

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2026-04-20

```
library(bio3d)
```

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```

```
include=FALSE  
knitr::opts_chunk$set(echo = TRUE)
```

```
# analyze_protein_bfactor()  
# Inputs:  
#   pdb_id - a Protein Data Bank ID as a character string (for example, "4AKE")  
#   chain   - the chain to analyze; default is "A"  
#   elety   - the atom type to keep; default is "CA" for alpha carbons  
#   typ      - the plot type passed to plotb3(); default is "l" for line plot  
#  
# What the function does:  
#   This function reads a PDB structure from the Protein Data Bank,  
#   trims it to a specified chain and atom type, extracts the B-factors,  
#   and plots the B-factor profile for that protein.  
#  
# Output:  
#   The function produces a B-factor plot and returns a list containing:  
#   1) the full PDB object  
#   2) the trimmed PDB object  
#   3) the extracted B-factor vector  
  
analyze_protein_bfactor <- function(pdb_id, chain = "A", elety = "CA", typ = "l") {  
  
  # Read the PDB structure  
  pdb <- read.pdb(pdb_id)
```

```

# Trim to the specified chain and atom type
pdb_trim <- trim.pdb(pdb, chain = chain, elety = elety)

# Error check in case no atoms are found
if (is.null(pdb_trim$atom) || nrow(pdb_trim$atom) == 0) {
  stop("No atoms found for the selected chain and atom type.")
}

# Extract B-factors
b_factors <- pdb_trim$atom$b

# Plot B-factors
plotb3(
  b_factors,
  sse = pdb_trim,
  typ = typ,
  ylab = "B-factor",
  main = paste("B-factor profile for", pdb_id, "chain", chain)
)

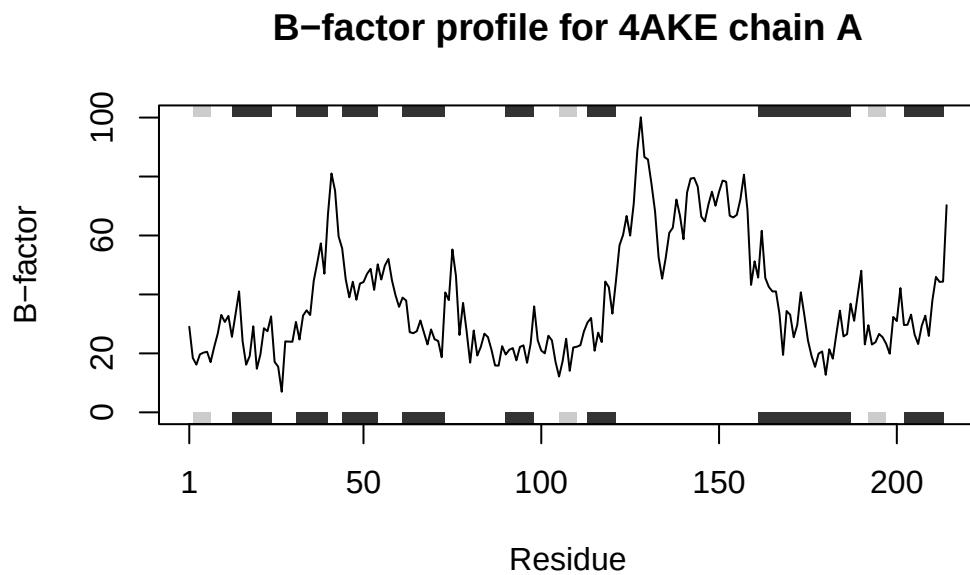
# Return useful output
return(list(
  pdb = pdb,
  trimmed_pdb = pdb_trim,
  b_factors = b_factors
))
}

```

First Example Code

```
result1 <- analyze_protein_bfactor("4AKE")
```

Note: Accessing on-line PDB file



I can also use the same function on other proteins from the assignment without rewriting the code each time.

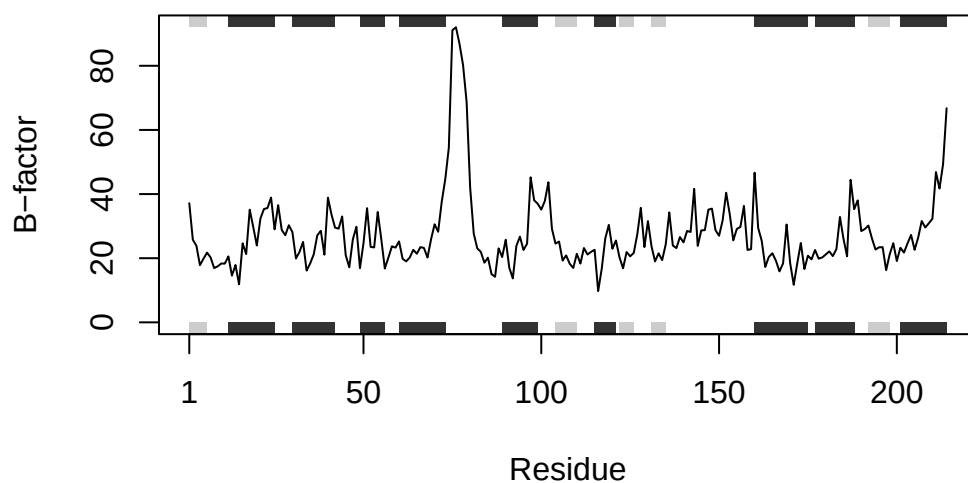
Additional example codes

```
result2 <- analyze_protein_bfactor("1AKE")
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE

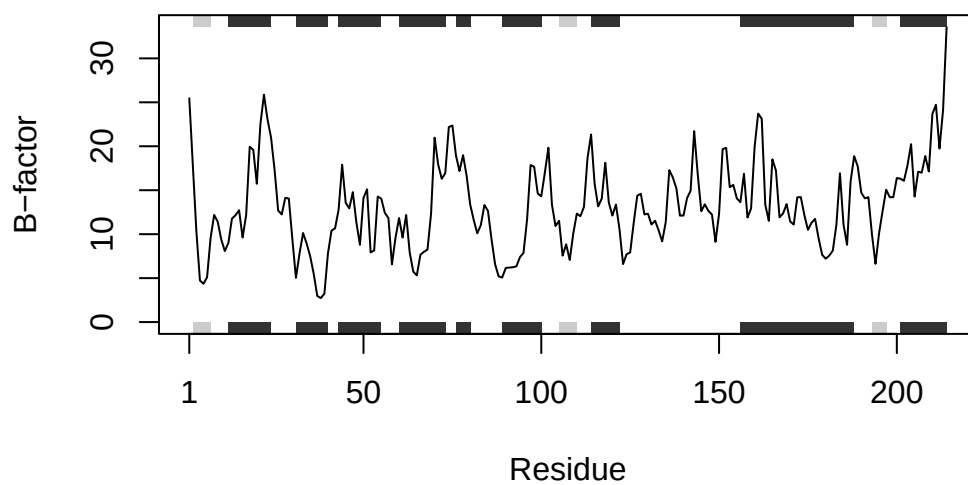
B-factor profile for 1AKE chain A



```
result3 <- analyze_protein_bfactor("1E4Y")
```

Note: Accessing on-line PDB file

B-factor profile for 1E4Y chain A



Discussion

I wrote the function “`analyze_protein_bfactor()`” to generalize the original repeated analysis code into one reusable function. Instead of manually reading, trimming, extracting B-factors, and plotting each protein separately, the function performs all of these steps automatically for any input PDB ID. This makes the code more efficient because it reduces duplication, improves readability through meaningful variable names and comments, and makes it easier to analyze multiple proteins. The function inputs are clearly documented, its purpose is explained, and its output is defined, which matches the rubric requirements for documentation and readable code. The function also returns useful objects for later analysis in addition to producing the plot. This follows the good function-writing practices described in the lecture notes